

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report
```

```
In [2]: iris = load_iris()
X=iris.data
y=(iris.target == 0).astype(int)
```

```
In [3]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
In [4]: scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
In [5]: model = tf.keras.Sequential([ tf.keras.layers.Dense(1, activation='sigmoid', input_shape=(X_train.shape[1],))])
```

C:\Users\subha\anaconda3\envs\tf_env\lib\site-packages\keras\src\layers\core\dense.py:93: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
In [6]: model.compile(optimizer='sgd', loss='binary_crossentropy', metrics=['accuracy'])
```

```
In [7]: history = model.fit(X_train, y_train, epochs=100, verbose=1)
```

Epoch 1/100
4/4  1s 28ms/step - accuracy: 0.8241 - loss: 0.4289
Epoch 2/100
4/4  0s 21ms/step - accuracy: 0.8512 - loss: 0.4030
Epoch 3/100
4/4  0s 19ms/step - accuracy: 0.8293 - loss: 0.4060
Epoch 4/100
4/4  0s 30ms/step - accuracy: 0.8523 - loss: 0.3900
Epoch 5/100
4/4  0s 18ms/step - accuracy: 0.8582 - loss: 0.4004
Epoch 6/100
4/4  0s 17ms/step - accuracy: 0.8491 - loss: 0.3929
Epoch 7/100
4/4  0s 16ms/step - accuracy: 0.8449 - loss: 0.4018
Epoch 8/100
4/4  0s 19ms/step - accuracy: 0.8675 - loss: 0.3734
Epoch 9/100
4/4  0s 19ms/step - accuracy: 0.8581 - loss: 0.3746
Epoch 10/100
4/4  0s 20ms/step - accuracy: 0.8831 - loss: 0.3610
Epoch 11/100
4/4  0s 25ms/step - accuracy: 0.8456 - loss: 0.3789
Epoch 12/100
4/4  0s 31ms/step - accuracy: 0.8581 - loss: 0.3536
Epoch 13/100
4/4  0s 19ms/step - accuracy: 0.8942 - loss: 0.3486
Epoch 14/100
4/4  0s 20ms/step - accuracy: 0.8724 - loss: 0.3487
Epoch 15/100
4/4  0s 17ms/step - accuracy: 0.8765 - loss: 0.3372
Epoch 16/100
4/4  0s 18ms/step - accuracy: 0.8838 - loss: 0.3376
Epoch 17/100
4/4  0s 17ms/step - accuracy: 0.8692 - loss: 0.3427
Epoch 18/100
4/4  0s 34ms/step - accuracy: 0.8671 - loss: 0.3412
Epoch 19/100
4/4  0s 20ms/step - accuracy: 0.8838 - loss: 0.3243
Epoch 20/100
4/4  0s 18ms/step - accuracy: 0.8668 - loss: 0.3242
Epoch 21/100

4/4  0s 20ms/step - accuracy: 0.8876 - loss: 0.3187
Epoch 22/100

4/4  0s 17ms/step - accuracy: 0.9001 - loss: 0.3055
Epoch 23/100

4/4  0s 18ms/step - accuracy: 0.8845 - loss: 0.3033
Epoch 24/100

4/4  0s 19ms/step - accuracy: 0.9067 - loss: 0.3012
Epoch 25/100

4/4  0s 22ms/step - accuracy: 0.9098 - loss: 0.3025
Epoch 26/100

4/4  0s 20ms/step - accuracy: 0.8963 - loss: 0.3007
Epoch 27/100

4/4  0s 19ms/step - accuracy: 0.9088 - loss: 0.2913
Epoch 28/100

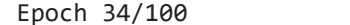
4/4  0s 19ms/step - accuracy: 0.9140 - loss: 0.2910
Epoch 29/100

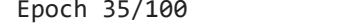
4/4  0s 17ms/step - accuracy: 0.8984 - loss: 0.2982
Epoch 30/100

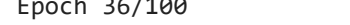
4/4  0s 21ms/step - accuracy: 0.9275 - loss: 0.2683
Epoch 31/100

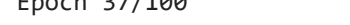
4/4  0s 17ms/step - accuracy: 0.8869 - loss: 0.2848
Epoch 32/100

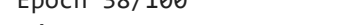
4/4  0s 16ms/step - accuracy: 0.9206 - loss: 0.2790
Epoch 33/100

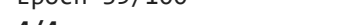
4/4  0s 21ms/step - accuracy: 0.9258 - loss: 0.2683
Epoch 34/100

4/4  0s 19ms/step - accuracy: 0.9060 - loss: 0.2904
Epoch 35/100

4/4  0s 18ms/step - accuracy: 0.9275 - loss: 0.2766
Epoch 36/100

4/4  0s 17ms/step - accuracy: 0.9296 - loss: 0.2707
Epoch 37/100

4/4  0s 16ms/step - accuracy: 0.9449 - loss: 0.2599
Epoch 38/100

4/4  0s 16ms/step - accuracy: 0.9455 - loss: 0.2605
Epoch 39/100

4/4  0s 18ms/step - accuracy: 0.9674 - loss: 0.2550
Epoch 40/100

4/4  0s 16ms/step - accuracy: 0.9612 - loss: 0.2441
Epoch 41/100

4/4  0s 17ms/step - accuracy: 0.9424 - loss: 0.2645

Epoch 42/100
4/4  0s 23ms/step - accuracy: 0.9660 - loss: 0.2530
Epoch 43/100
4/4  0s 21ms/step - accuracy: 0.9546 - loss: 0.2583
Epoch 44/100
4/4  0s 18ms/step - accuracy: 0.9639 - loss: 0.2486
Epoch 45/100
4/4  0s 17ms/step - accuracy: 0.9671 - loss: 0.2399
Epoch 46/100
4/4  0s 18ms/step - accuracy: 0.9452 - loss: 0.2522
Epoch 47/100
4/4  0s 21ms/step - accuracy: 0.9813 - loss: 0.2415
Epoch 48/100
4/4  0s 18ms/step - accuracy: 0.9792 - loss: 0.2298
Epoch 49/100
4/4  0s 17ms/step - accuracy: 0.9542 - loss: 0.2380
Epoch 50/100
4/4  0s 17ms/step - accuracy: 0.9604 - loss: 0.2395
Epoch 51/100
4/4  0s 28ms/step - accuracy: 0.9542 - loss: 0.2431
Epoch 52/100
4/4  0s 19ms/step - accuracy: 0.9698 - loss: 0.2232
Epoch 53/100
4/4  0s 16ms/step - accuracy: 0.9636 - loss: 0.2283
Epoch 54/100
4/4  0s 17ms/step - accuracy: 0.9698 - loss: 0.2263
Epoch 55/100
4/4  0s 17ms/step - accuracy: 0.9604 - loss: 0.2167
Epoch 56/100
4/4  0s 17ms/step - accuracy: 0.9813 - loss: 0.2043
Epoch 57/100
4/4  0s 18ms/step - accuracy: 0.9698 - loss: 0.2083
Epoch 58/100
4/4  0s 17ms/step - accuracy: 0.9872 - loss: 0.2194
Epoch 59/100
4/4  0s 17ms/step - accuracy: 0.9757 - loss: 0.2105
Epoch 60/100
4/4  0s 25ms/step - accuracy: 0.9757 - loss: 0.2237
Epoch 61/100
4/4  0s 16ms/step - accuracy: 0.9788 - loss: 0.2190
Epoch 62/100

4/4  0s 17ms/step - accuracy: 0.9757 - loss: 0.2001
Epoch 63/100

4/4  0s 20ms/step - accuracy: 0.9882 - loss: 0.1942
Epoch 64/100

4/4  0s 19ms/step - accuracy: 0.9757 - loss: 0.2030
Epoch 65/100

4/4  0s 16ms/step - accuracy: 0.9757 - loss: 0.2143
Epoch 66/100

4/4  0s 18ms/step - accuracy: 0.9847 - loss: 0.1969
Epoch 67/100

4/4  0s 16ms/step - accuracy: 0.9941 - loss: 0.1871
Epoch 68/100

4/4  0s 18ms/step - accuracy: 0.9941 - loss: 0.1912
Epoch 69/100

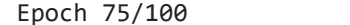
4/4  0s 26ms/step - accuracy: 1.0000 - loss: 0.2021
Epoch 70/100

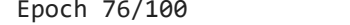
4/4  0s 21ms/step - accuracy: 1.0000 - loss: 0.1937
Epoch 71/100

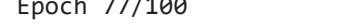
4/4  0s 27ms/step - accuracy: 1.0000 - loss: 0.1925
Epoch 72/100

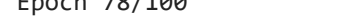
4/4  0s 22ms/step - accuracy: 1.0000 - loss: 0.2017
Epoch 73/100

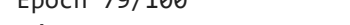
4/4  0s 16ms/step - accuracy: 1.0000 - loss: 0.1952
Epoch 74/100

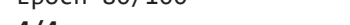
4/4  0s 22ms/step - accuracy: 1.0000 - loss: 0.1835
Epoch 75/100

4/4  0s 18ms/step - accuracy: 1.0000 - loss: 0.1826
Epoch 76/100

4/4  0s 18ms/step - accuracy: 1.0000 - loss: 0.1907
Epoch 77/100

4/4  0s 18ms/step - accuracy: 1.0000 - loss: 0.1777
Epoch 78/100

4/4  0s 19ms/step - accuracy: 1.0000 - loss: 0.1888
Epoch 79/100

4/4  0s 17ms/step - accuracy: 1.0000 - loss: 0.1891
Epoch 80/100

4/4  0s 19ms/step - accuracy: 1.0000 - loss: 0.1797
Epoch 81/100

4/4  0s 20ms/step - accuracy: 1.0000 - loss: 0.1817
Epoch 82/100

4/4  0s 27ms/step - accuracy: 1.0000 - loss: 0.1812

Epoch 83/100
4/4 ————— 0s 31ms/step - accuracy: 1.0000 - loss: 0.1750
Epoch 84/100
4/4 ————— 0s 17ms/step - accuracy: 1.0000 - loss: 0.1767
Epoch 85/100
4/4 ————— 0s 16ms/step - accuracy: 1.0000 - loss: 0.1725
Epoch 86/100
4/4 ————— 0s 20ms/step - accuracy: 1.0000 - loss: 0.1695
Epoch 87/100
4/4 ————— 0s 16ms/step - accuracy: 1.0000 - loss: 0.1697
Epoch 88/100
4/4 ————— 0s 17ms/step - accuracy: 1.0000 - loss: 0.1887
Epoch 89/100
4/4 ————— 0s 23ms/step - accuracy: 1.0000 - loss: 0.1687
Epoch 90/100
4/4 ————— 0s 17ms/step - accuracy: 1.0000 - loss: 0.1651
Epoch 91/100
4/4 ————— 0s 20ms/step - accuracy: 1.0000 - loss: 0.1638
Epoch 92/100
4/4 ————— 0s 18ms/step - accuracy: 1.0000 - loss: 0.1683
Epoch 93/100
4/4 ————— 0s 24ms/step - accuracy: 1.0000 - loss: 0.1596
Epoch 94/100
4/4 ————— 0s 19ms/step - accuracy: 1.0000 - loss: 0.1700
Epoch 95/100
4/4 ————— 0s 18ms/step - accuracy: 1.0000 - loss: 0.1675
Epoch 96/100
4/4 ————— 0s 16ms/step - accuracy: 1.0000 - loss: 0.1587
Epoch 97/100
4/4 ————— 0s 25ms/step - accuracy: 1.0000 - loss: 0.1706
Epoch 98/100
4/4 ————— 0s 16ms/step - accuracy: 1.0000 - loss: 0.1659
Epoch 99/100
4/4 ————— 0s 18ms/step - accuracy: 1.0000 - loss: 0.1758
Epoch 100/100
4/4 ————— 0s 17ms/step - accuracy: 1.0000 - loss: 0.1552

```
In [8]: predictions = model.predict(X_test)
        binary_predictions = (predictions > 0.5).astype(int)
```

2/2 ————— 0s 73ms/step

```
In [9]: class_names = ['Non-Setosa', 'Setosa']
report = classification_report(y_test, binary_predictions, target_names=class_names)
```

```
In [10]: print("Classification Report:\n")
print(report)
```

Classification Report:

	precision	recall	f1-score	support
Non-Setosa	1.00	1.00	1.00	26
Setosa	1.00	1.00	1.00	19
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

```
In [11]: plt.figure(figsize=(8, 6))
plt.scatter(X_test[:, 0], X_test[:, 1], c=y_test, cmap='coolwarm', marker='o', label='True Class')
plt.xlabel('Sepal Length')
plt.ylabel('Sepal Width')
x_vals = np.linspace(X_test[:, 0].min(), X_test[:, 0].max(), 100)
y_vals = (-model.weights[0][0] * x_vals - model.weights[1]) / model.weights[0][1]
plt.plot(x_vals, y_vals, 'r--', label='Decision Boundary')
plt.legend()
plt.show()
```

